TASK SPECIFICATION: P5-08 PHASE P5 — SMART PARKING & ALPR COMPUTER VISION GATE

Flex-Share Marketplace UI

Git Branch: `feature/P5-08-flex-ui` · Target Pull Request: feature/P5-08-flex-ui

Core Objective & Technical Scope:
One-click spot sharing activation toggle, projected reward estimator, and active exchange status tracker.

Priority Level	Medium	Complexity Rating	`M`
Execution Status	Backlog 📋	Target PR / Commit	`feature/P5-08-flex-ui`
Upstream Dependencies	P5-04, P5-06		
Architecture Blockers	None (Ready for execution)		

IMPLEMENTATION ROADMAP & ENGINEERING STEPS

1. Build FlexShareToggle and reward estimation widget.
2. Build active exchange status banner and claim confirmation dialog.
3. Wire components to Flex-Share API endpoints.
4. Handle expiration and cancellation states.

TARGET FILE INVENTORY & EXPECTED ARTIFACTS

- `frontend/src/features/parking/flex/*`

ACCEPTANCE CRITERIA & QUALITY CHECKLIST

- Exact visual match with master design showcase.
- Prevents duplicate offer activation on already shared spots.

VERIFICATION & TEST COVERAGE STRATEGY

- FlexShare component tests
- Playwright exchange flow